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RESEARCH ARTICLE

Temporal-Aware Transformer Approach for Violence Activity Recognition

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ABSTRACT The need for effective violence detection in public spaces has intensified with increasing antisocial behavior and violence. Traditional surveillance systems, which are relying on human operators, face delays and resource challenges. Using advances in artificial intelligence (AI) and computer vision, this research presents a scalable deep learning architecture for real-time violence detection using two approaches. In the first approach, Convolutional Neural Networks (CNN) and bidirectional long-short-term memory (BiLSTM) networks are combined, where MobileNetV2 is used for spatial feature extraction and BiLSTM for temporal pattern recognition, achieving an accuracy of 95.6%. The second approach incorporates a spatial-temporal transformer (TransformerSeq) in place of BiLSTM, improving performance to 97.2% by capturing spatiotemporal relationships in video data more effectively through self-attention for temporal feature learning. The lightweight SOTA MobileNetV2, along with the proposed MobileTransformerSeq, enables the effective differentiation between violent and non-violent activities, demonstrating the potential to enhance public safety in diverse settings.

INDEX TERMS Deep learning, MobileNetV2, self-attention, Transformer, violence detection.

I. INTRODUCTION

Violence, vandalism, and other antisocial activities in public spaces are recognized as increasingly important concerns in both urban and rural environments, affecting the general well-being and security of communities. Despite the severity of these problems, traditional measures for preventing or controlling violence often prove ineffective, as they rely on human intervention, which remains subject to delays and inefficiencies [1]. The complexity of these problems is further compounded by underlying socio-political and economic factors that contribute to their occurrence [2], [3], [4], [5], [6]. Although complete eradication of such acts would require advanced and invasive methods, such as real-time monitoring of brain signals [7], [8] to understand the root

causes of violent impulses, such an approach remains neither technically feasible nor scalable in real-world settings.

Recent advancements in artificial intelligence, particularly in computer vision and video analytics, have created new opportunities for addressing the challenges of violence detection. With surveillance systems already integrated into public spaces such as streets, parks, transit hubs, and workplaces, the necessary infrastructure exists for a more robust and automated approach. AI-driven models can analyze real-time footage from these cameras, identifying suspicious or violent behavior as it unfolds, thereby reducing the need for constant human monitoring and enabling quicker response times.

The integration of AI into surveillance systems enhances the ability to intervene rapidly during violent incidents, preventing escalation and minimizing harm to individuals and property. A key aspect of this advancement is the adoption of lightweight architectures in neural networks,

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which are increasingly relevant in Transformer models for deployment on resource-constrained devices and edge computing platforms. To strengthen this perspective, we have incorporated a dedicated discussion on the significance of lightweight architectures in the introduction, supported by the reference [9] and other relevant citations. These modifications provide a stronger theoretical foundation for our work and align it with recent progress in efficient deep learning architectures.

Upon detecting a violent act, an AI-powered surveillance system can trigger immediate alerts to law enforcement or security personnel, allowing for rapid intervention before the situation worsens. The deployment of machine learning models to recognize patterns of violent behavior adds a proactive layer of intelligence to existing security frameworks, making them more responsive and effective.

Beyond public spaces, the potential applications of this technology extend to critical environments such as schools, hospitals, and other institutions where safety is a priority. AI-powered smart surveillance not only strengthens security measures but also fosters a sense of safety by ensuring that acts of violence are detected and addressed in real time. Ongoing research in this field aims to refine AI-based models, improving their accuracy, detection speed, and robustness across diverse real-world conditions, including variations in lighting, crowded environments, and different forms of human interaction.

Ultimately, the objective is to create safer public environments by leveraging technology that operates autonomously, significantly reducing the burden on human resources while improving response times. By emphasizing real-life violence detection through AI and video analytics, this research contributes to a future in which society can more effectively prevent and manage antisocial activities, ensuring greater security in public spaces.

A. NOVELTY

The novelty of the proposed model lies in its unique hybrid architecture, which seamlessly integrates multiple deep learning components to enhance violence detection accuracy and efficiency. MobileNetV2 is used to extract spatial features from individual video frames, ensuring that essential structural and contextual details are captured while maintaining computational efficiency. This allows the model to process real-time data effectively, making it suitable for deployment on resource-constrained devices, such as surveillance cameras and embedded systems.

To capture temporal dependencies across video frames, Bidirectional Long Short-Term Memory (BiLSTM) is incorporated, enabling the model to analyze both past and future contextual information. This bidirectional processing enhances the recognition of violent activities that may develop over time, rather than relying solely on instantaneous frame analysis. However, recurrent architectures like

BiLSTM may struggle with long-range dependencies and complex temporal relationships, which require the inclusion of TransformerSeq.

TransformerSeq, equipped with a self-attention mechanism, is employed to model intricate frame-to-frame relationships by identifying significant patterns across the entire video sequence. Unlike traditional models that process sequential data in a linear manner, TransformerSeq enables parallel processing, thereby improving efficiency and the ability to detect subtle variations in movement indicative of violent actions.

By integrating these three components—MobileNetV2 for spatial feature extraction, BiLSTM for sequential pattern recognition, and TransformerSeq for advanced temporal modeling—this dual-stream architecture delivers a scalable and high-performance solution. The combined approach ensures robustness against challenging conditions such as poor lighting, occlusions, and crowded environments, where single-dimensional feature analysis often fails. As a result, the proposed model surpasses conventional methods by offering a more comprehensive and intelligent framework for real-time violence detection in diverse real-world scenarios.

II. RELATED WORKS

Real-time violence detection has experienced significant advancements in recent years, driven by the need for better public safety and more efficient surveillance systems. Early methodologies in this domain are mainly based on handmade features and traditional machine learning techniques, which, although effective to some extent, lacked the capability to handle complex and dynamic video data. With the evolution of deep learning techniques, research has shifted towards models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), enabling the simultaneous extraction of spatial and temporal information.

Several notable studies have explored the use of CNNs and LSTMs for the detection of violence. A hybrid CNN-LSTM model has been proposed by Orozco et al. [10] for real-time surveillance-based violence detection. Al-Faris et al. analyzed the effectiveness of CNN-based architectures in human action recognition, emphasizing their robustness in handling occlusions and varying lighting conditions in surveillance footage [11].

Singh et al. introduced a drone-based system that uses deep neural networks to detect violent individuals, focusing on real-time applications where rapid threat identification is crucial [12]. A similar approach had been adopted by Dhruv Shinde et al., who integrated CNN models with CCTV cameras to monitor public spaces, achieving high accuracy in detecting physical altercations [13].

Recent studies have expanded the scope to lightweight architectures, such as YOLO, which have demonstrated both computational efficiency and high accuracy. Zhang et al. employed YOLO-based methods to track and detect violent actions in crowded environments [14]. These methods capitalize on YOLO's ability to detect objects in real time

with minimal latency, making them suitable for large public gatherings.

Beyond CNNs, optical flow methods have been utilized to enhance the temporal detection of violent movements. Cao et al. incorporated optical flow with deep learning architectures to identify violent actions by tracking motion patterns in sequential video frames [15]. Soomro et al. have used the UCF101 dataset for training, which remains a standard benchmark for human action recognition tasks [16].

An additional contribution has been made by Dhruv Shindhe et al., who have integrated CNNs with attention mechanisms to improve model focus on critical frames exhibiting violent behavior [13]. Attention mechanisms have recently gained prominence in real-time violence detection, as they improve model efficiency by selectively processing the most relevant parts of the input video, thereby reducing computational costs.

Moreover, transfer learning has been extensively explored. Laptev et al. demonstrated the effectiveness of pre-trained models on large video datasets such as HMDB51, which encompass a wide range of violent and non-violent human activities [17]. This approach has facilitated faster training and improved generalization in diverse environments.

Despite these advances, several challenges persist in real-time violence detection. Occlusion, low video quality, and lighting variations continue to affect performance in real-world scenarios, as observed by Kuehne et al. in their study on recognition of human actions [18]. Future research will likely focus on addressing these limitations while enhancing the computational feasibility of violence detection models for deployment on edge devices such as drones and body cameras.

III. PROPOSED MODEL

The proposed system is trained and evaluated on a comprehensive dataset, utilizing a standard train-test split to assess its generalization capability. Designed for real-time video processing, the system ensures scalability and adaptability across various real-world applications. The methodological workflow of the proposed system is illustrated in Figure 1.

A. DATASET

To address the scarcity of video data labeled related to violence, a publicly accessible dataset developed by [19] has been utilized for this study. The dataset comprises 2,000 videos collected from YouTube, evenly distributed into 1,000 violent and 1,000 non-violent samples. Inclusion of real-life street fight scenarios in various environments and lighting conditions enhances the model's ability to generalize across diverse real-world situations.

The non-violent category includes videos depicting a range of human activities such as sports, eating, walking, and other daily actions. This diverse selection provides a robust negative class, ensuring that the model effectively differentiates between violent and non-violent behaviors.

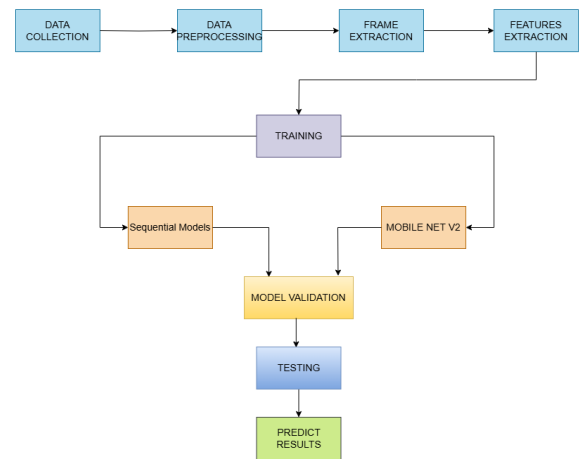


FIGURE 1. Flowchart of the proposed methodology.

This dataset has been developed in response to the limited availability of extensive video datasets specifically focused on interpersonal violence. By incorporating varied scenes and settings, it serves as a valuable resource for training deep learning models in violence recognition.

B. DATASET PREPARATION

To train the model, a curated dataset, as described earlier, has been utilized. This dataset consists of video clips explicitly labeled as “Violence” or “Non-Violence” to ensure precise classification. The dataset has been structured to incorporate diverse scenarios, including various forms of violence (e.g., physical altercations, aggressive behaviors) and non-violent activities (e.g., sports, everyday interactions). All videos have been organized into separate folders according to their respective labels. Additionally, multiple perspectives, environments, and lighting conditions are included to enhance the model's robustness.

C. DATA PREPROCESSING

1) VIDEO EXTRACTION

Each video clip is processed to extract individual frames using OpenCV, allowing for precise frame retrieval.

2) FRAME SAMPLING

To maintain uniform input sequences, a frame-skipping technique is applied, where every second frame is omitted to improve computational efficiency. The total frame count of each video is divided by a predefined sequence length (16), ensuring that frames are sampled at regular intervals. This technique effectively captures temporal dynamics while reducing computational complexity.

3) FRAME RESIZING

Extracted frames are resized to fixed dimensions ($IMAGE - HEIGHT \times IMAGE - WIDTH$)(64 × 64) to standardize

input data. This ensures that all frames are of consistent size, allowing the CNN to process them effectively.

4) NORMALIZATION

To enhance model training, the pixel values of the frames are normalized to the range [0, 1] by dividing by 255. This step accelerates convergence and stabilizes the learning process.

5) LABEL ENCODING

Each video's label is encoded in a suitable format for model training. For binary classification, labels are assigned integer values: **0** for “Non-Violence” and **1** for “Violence”.

D. DATA PREPARATION

The extracted frames and corresponding labels are organized using NumPy arrays. The dataset is then split into training and testing sets using Scikit-learn's `train_test_split` function, allocating **80% for training, 10% for validation, and 10% for testing**. One-hot encoding is applied to the labels to facilitate categorical classification.

E. MODEL ARCHITECTURE

Our proposed model leverages the **MobileNetV2** backbone for lightweight feature extraction, followed by a **BiLSTM** module to capture temporal dependencies across frames. This architecture provides an efficient solution suitable for real-time applications while maintaining a high level of accuracy in violence detection.

Furthermore, our approach integrates a hybrid **transformer-based sequence** processing mechanism to analyze temporal patterns in video frames for precise detection of violence. The architecture consists of three key modules: **Frame Sequence Encoding**, **Key Region Extraction**, and **Classification**. Each module is designed to effectively capture spatial and temporal features of video sequences. The schematic of the proposed model is shown in Figure 2.

1) MOBILENETV2

MobileNetV2 is a lightweight CNN architecture designed for mobile and embedded vision applications, as shown in Figure 3 [20]. Unlike traditional CNN architectures such as VGG or ResNet, MobileNetV2 introduces separable convolutions in-depth, which significantly reduce the computational cost and the number of parameters [21]. This makes MobileNetV2 particularly suitable for real-time tasks, such as detecting violence, where computational efficiency is paramount [22]. In this model, MobileNetV2 is employed to extract spatial features from individual video frames, allowing the capture of hierarchical patterns such as textures, edges, and shapes [23].

The key components of MobileNetV2 include the following:

- **Inverted Residuals and Linear Bottlenecks:** The architecture features an inverted residual structure with

linear bottlenecks, which allows the model to retain essential low-dimensional features while minimizing information loss during depthwise convolutions [20].

- **ReLU6 Activations:** ReLU6 activations are used to introduce non-linearities, offering improved performance in low-precision computing environments [21].

Using MobileNetV2, the model ensures efficient extraction of relevant spatial features, even in resource-constrained environments such as real-time video processing [24].

2) BIDIRECTIONAL LSTM

While CNNs excel at capturing spatial information, recognizing temporal dependencies within video sequences is crucial to detect violent behavior, which often involves sequential and dynamic motion patterns. To address this, a **Bidirectional LSTM (BiLSTM)** is employed.

The BiLSTM process video frames in both forward and backward directions, capturing both past and future context within the sequence [25]. This bidirectional processing proves to be especially beneficial for violence detection, as it enables the model to learn dependencies that span several frames [26]. For example, a single violent action may involve multiple subtle cues spread over time, and these can only be detected by considering future frames alongside past ones [27].

Figure 4 shows a simple LSTM cell, while Figure 5 presents the structure of a Bidirectional LSTM. Key components of the BiLSTM include:

- **Forget Gates:** These gates allow the model to selectively retain or discard information from previous frames, enabling it to focus on the most relevant temporal patterns [25].
- **Bidirectional Processing:** By processing the sequence in both directions, the model captures temporal dynamics more comprehensively. This is particularly important for complex actions such as violence, where future actions may provide context for previous movements [28].

Mathematically, let ℓ represent the number of hidden units, b the batch size, and f the number of input features. The hidden state from the previous time step is denoted as $S_{t-1} \in \mathbb{R}^{b \times \ell}$, and the input in the current time step is $Y_t \in \mathbb{R}^{b \times f}$. At time step t , the following gates are defined in Eq. 1:

$$P_t^* \in \mathbb{R}^{b \times \ell}, \quad Q_t^* \in \mathbb{R}^{b \times \ell}, \quad R_t^* \in \mathbb{R}^{b \times \ell} \quad (1)$$

These gates correspond to the input, forget, and output operations, respectively. Their computation is described as follows in Eqs. 2, 3 and 4:

$$P_t^* = \sigma(U_{yp}Y_t'' + U_{sp}S_{t-1} + \beta_p) \quad (2)$$

$$Q_t^* = \sigma(U_{yq}Y_t'' + U_{sq}S_{t-1} + \beta_q) \quad (3)$$

$$R_t^* = \sigma(U_{yr}Y_t'' + U_{sr}S_{t-1} + \beta_r) \quad (4)$$

In these equations, $U_{yp}, U_{yq}, U_{yr} \in \mathbb{R}^{f \times \ell}$ and $U_{sp}, U_{sq}, U_{sr} \in \mathbb{R}^{\ell \times \ell}$ represent weight parameters, while $\beta_p, \beta_q, \beta_r \in \mathbb{R}^{1 \times \ell}$ are bias parameters.

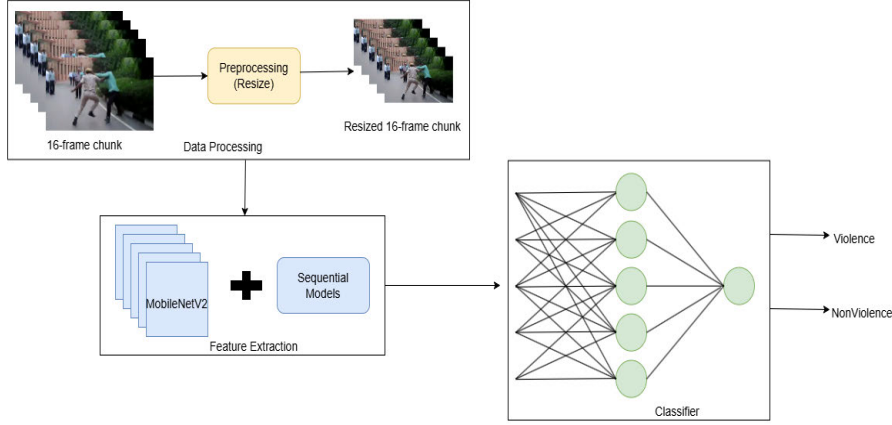


FIGURE 2. Visualization of frame-wise classification between Non-Violence and Violence.

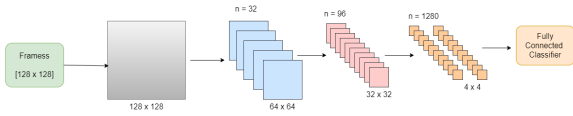


FIGURE 3. Frame segmentation using MobileNetV2.

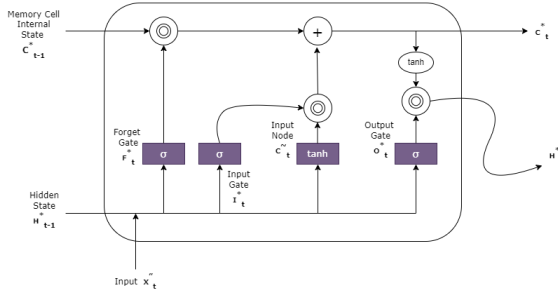


FIGURE 4. Structure of a simple LSTM cell.

In further detail, an auxiliary input node $\tilde{Z}_t \in \mathbb{R}^{b \times \ell}$ is introduced. It is calculated at the time step t using the Eq. 5:

$$\tilde{Z}_t = \tanh(Y_t'' U_{yz} + S_{t-1}^* U_{sz} + \beta_z) \quad (5)$$

where $\beta_z \in \mathbb{R}^{1 \times \ell}$ is a bias term and $U_{yz} \in \mathbb{R}^{f \times \ell}$, $U_{sz} \in \mathbb{R}^{\ell \times \ell}$ are weight matrices.

The input gate P_t manages how much new information to incorporate using $\tilde{Z}_t$ and the forget gate. Retention of the previous cell state ($Z_{t-1} \in \mathbb{R}^{b \times \ell}$) is governed by Q_t . The updated cell state is computed using the Hadamard product (element-wise) as follows in Eq. 6:

$$Z_t^* = Q_t^* \odot Z_{t-1}^* + P_t^* \odot \tilde{Z}_t \quad (6)$$

Finally, the hidden state $S_t \in \mathbb{R}^{b \times \ell}$, which serves as the output of the memory cell, is computed using the output gate. The internal state undergoes activation $\tanh$, followed by

point-wise multiplication with the output gate. This ensures that the hidden state values remain within $(-1, 1)$ (see Eq. 7):

$$S_t^* = R_t^* \odot \tanh(Z_t^*) \quad (7)$$

Integrating MobileNetV2 with BiLSTM enables the model to efficiently capture spatial and temporal characteristics, facilitating the accurate detection of violent actions in real-time video streams [29].

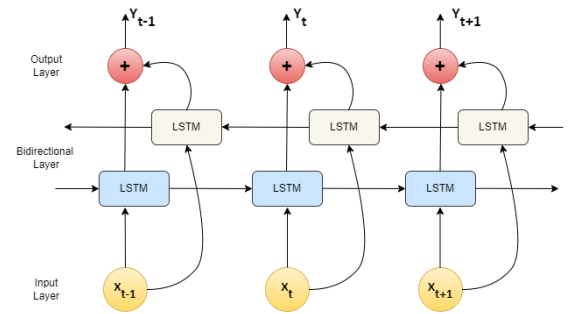


FIGURE 5. Structure of a bidirectional LSTM.

3) TRANSFORMERSEQ

As shown in Figure 6, the **TransformerSeq** model is designed to capture the temporal connections between video frames, which are essential for tasks such as real-time violence detection [30]. By utilizing the self-attention mechanism of Transformers to focus on frame dependencies, TransformerSeq captures intricate temporal patterns in the video sequence. This is in contrast to conventional sequential models, such as LSTMs, which rely on recurrent mechanisms [31].

1) Positional Encoding

Since the Transformer models do not intrinsically recognize the order of sequences, an attainable positional

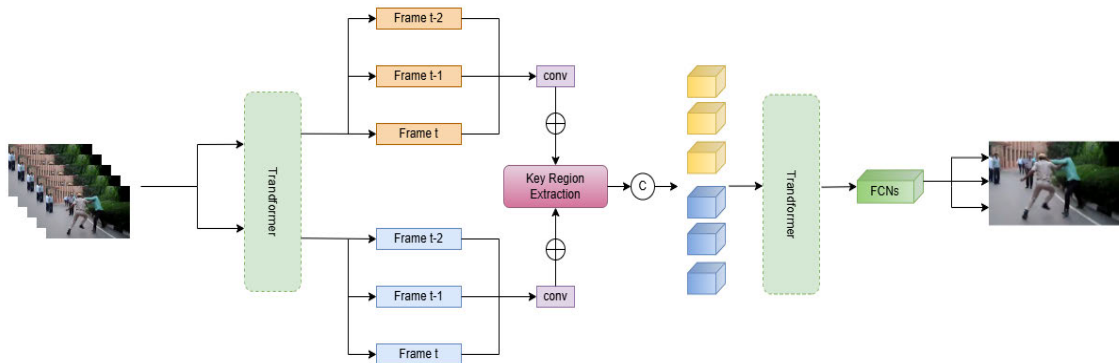


FIGURE 6. Sequencing by transformer used in our Proposed MobileNetV2-TransformerSeq model.

encoding is used in TransformerSeq. The positional encoding for a given position p and dimension k is defined as Eqs. 8 and 9:

$$PE_{(p,2k)} = \sin\left(\frac{p}{10000^{2k/m}}\right) \quad (8)$$

$$PE_{(p,2k+1)} = \cos\left(\frac{p}{10000^{2k/m}}\right) \quad (9)$$

In this formulation, $PE_{(p,2k)}$ and $PE_{(p,2k+1)}$ denote the sine and cosine positional encodings. The term $10000^{2k/m}$ scales the position p relative to the embedding dimension m .

2) Transformer Encoder Layers

The heart of the TransformerSeq model is its **Transformer encoder layers**, which employ multi-head self-attention to identify important relationships between frames. These encoder layers consist of the following key components:

- **Self-Attention Mechanism:** The self-attention mechanism calculates the attention scores as follows (see in Eq. 10):

$$\text{Attention}(A, B, C) = \text{softmax}\left(\frac{AB^T}{\sqrt{d_b}}\right)C \quad (10)$$

In this equation, A , B , and C represent the query, key, and value matrices, respectively, while d_b denotes the dimensionality of the key vectors. This mechanism enables the model to focus on the most significant frames within the sequence, effectively learning the interdependencies and capturing long-term temporal patterns essential for detecting violent activities.

- **Feedforward Layers:** Following the self-attention mechanism, feedforward layers are employed to refine the captured temporal dependencies. The feedforward operation is defined as Eq. 11:

$$\text{FFN}(z) = \text{ReLU}(zU_1 + c_1)U_2 + c_2 \quad (11)$$

In this expression, U_1 and U_2 are learnable weight matrices, c_1 and c_2 are bias terms, and z represents the input feature vector.

3) Classification Layer

Once the temporal dependencies are captured by the Transformer encoder, the output corresponding to the last frame in the sequence is passed through a fully connected classification layer. This layer consolidates the temporal features learned by the Transformer model and classifies the video sequence as either violent or non-violent. The classification is defined as Eq. 12:

$$\tilde{y} = \text{softmax}(vU_c + c_c) \quad (12)$$

Here, v represents the feature vector output by the Transformer encoder, U_c is the weight matrix, and c_c is the bias vector.

4) Advantages of TransformerSeq:

- **Effective Temporal Representation:** By leveraging self-attention, TransformerSeq can focus on significant frames in a sequence, capturing subtle temporal patterns indicative of violent actions.
- **Scalability and Efficiency:** TransformerSeq's lightweight structure enables effective processing, which makes it appropriate for real-time applications like surveillance systems.

In conclusion, by successfully simulating temporal dynamics across video frames using self-attention, a critical component for real-time violence detection, the TransformerSeq model greatly improves the system's capacity to detect violence.

F. EVALUATION

After training, the model performance is assessed using the test dataset. Together, the evaluation metrics of accuracy, precision, recall, and the confusion matrix provide insight into the model's ability to discriminate between violent and non-violent content.

The model demonstrated its effectiveness in differentiating between violent and non-violent content with an overall

accuracy of 94% in the test dataset, precision and recall scores of 96% and 91% in the non-violence dataset, and precision and recall scores of 92% and 96% in the violence dataset, respectively.

1) REAL-TIME PREDICTION

The predict-video function is implemented to analyze new video inputs:

- **Frame Extraction and Preprocessing:** This function extracts frames from the specified video and applies the same preprocessing steps that had been used during training.
- **Model Inference:** Class probabilities are obtained by running the preprocessed frames through the trained model. The class with the highest probability is selected to determine the predicted class.
- **Confidence Display:** The predicted class name and its confidence score are displayed, indicating the reliability of the prediction.
- **Video Visualization:** The function allows for the playback of the original video along with the prediction results, facilitating the qualitative evaluation of the model's performance.

IV. RESULT AND ANALYSIS

The results of the violence recognition model developed in this study using the dataset mentioned in the preceding sections are presented in this section. A detailed analysis of the evaluation metrics, classification report, confusion matrix, and performance visualizations is conducted to identify the advantages and limitations of the model. The model's resilience and generalizability are further assessed using accuracy, precision, recall, and F1 score. The results are contextualized through qualitative visualizations of model predictions and comparisons with existing state-of-the-art models.

A. MODEL PERFORMANCE

A balanced dataset consisting of 1000 violent and 1000 non-violent video examples, which is used to train the proposed MobileTransformerSeq model (MobileNetV2 + TransformerSeq). Ten percent of the dataset has been used for testing, ten percent for validation, and eighty percent for training. This train-test split, with a distinct validation set aside during training and a separate test set for final evaluation, ensures that the model has been trained on diverse data.

Specifically, 1600 videos are used for testing, 200 for validation, and 200 for training. The validation set provided insights into the model's performance during training, assisting in monitoring and fine-tuning the learning process, while the training set enabled the model to learn patterns that differentiate violent from non-violent videos. The test set, which had not been used during training or validation, had been used to assess the model's ability to generalize to new, unseen data.

After 15 epochs of training with a batch size of 8 and a validation split of 20%, the performance of the model is evaluated in the test set. Evaluation metrics included F1 score, recall, accuracy, and precision for both violent and non-violent classes.

The model exhibited strong generalization to unseen data, achieving an overall test accuracy of 97.2%. The steady convergence observed in the training and validation dynamics supports this performance.

Furthermore, the MobBiLSTM (MobileNetV2 + BiLSTM) model has been trained using 1000 violent and 1000 non-violent video clips. The dataset has been divided into 80% for training, 10% for validation, and 10% for testing to ensure fair evaluation.

The MobBiLSTM model demonstrated its ability to detect violent actions with an overall test accuracy of 95.6%. Stable training and validation dynamics indicated that the model effectively captured spatio-temporal information without overfitting.

1) TRAINING AND LOSS AND VALIDATION ACCURACY

The training and validation accuracy over epochs is shown in Figure 7, while Figure 8 illustrates the corresponding loss values. These curves provide insight into the learning processes of both models:

- **Training & Validation Accuracy:** Both the training and validation accuracy improved steadily during the epochs, with minimal divergence between the MobileTransformerSeq and MobBiLSTM models, as seen in Figure 7. This suggests that both models successfully identified patterns in the data without overfitting.
- **Training & Validation Loss:** The loss values for both training and validation sets consistently decreased over epochs for both models, as shown in Figure 8. The absence of significant spikes in loss values reflects the trends in accuracy and confirms the stability of the learning process.

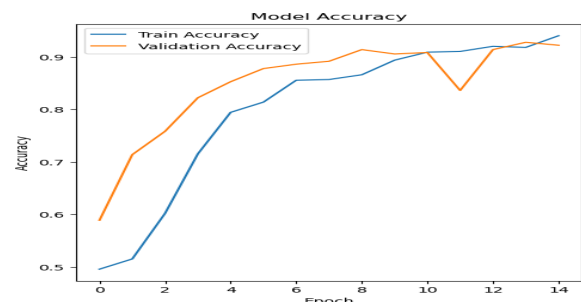


FIGURE 7. Training and validation accuracy.

B. ACCURACY AND CLASSIFICATION PERFORMANCE

After training the models, the overall accuracy reaches 97.2% for MobileTransformerSeq and 95.6% for MobBiLSTM, indicating their efficacy in distinguishing between violence

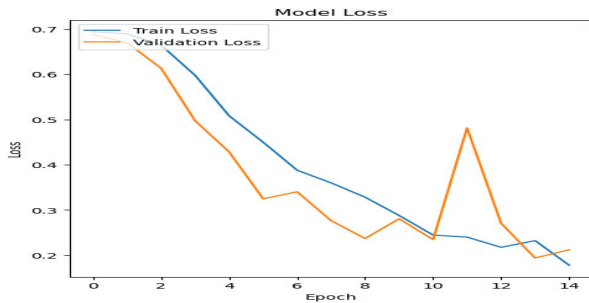


FIGURE 8. Training and validation loss.

and non-violence in video data. The detailed classification reports for both models are presented below, offering a breakdown of performance for each class, i.e., violence (class 1) and non-violence (class 0):

TABLE 1. Classification report of proposed MobileTransformerSeq model.

Class	Precision	Recall	F1-score	Support
Non-Violence (0)	0.98	0.96	0.97	97
Violence (1)	0.96	0.98	0.97	103
Accuracy			0.97	200
Macro avg	0.97	0.97	0.97	200
Weighted avg	0.97	0.97	0.97	200

TABLE 2. Classification report of proposed MobBiLSTM model.

Class	Precision	Recall	F1-score	Support
Non-Violence (0)	0.96	0.91	0.93	99
Violence (1)	0.92	0.96	0.94	101
Accuracy			0.95	200
Macro avg	0.94	0.93	0.93	200
Weighted avg	0.94	0.94	0.93	200

- **Precision:** MobileTransformerSeq achieves a precision of 0.98 for non-violence and 0.96 for violence, indicating slightly higher precision in predicting non-violence. MobBiLSTM achieves 0.96 for non-violence and 0.92 for violence.
- **Recall:** MobileTransformerSeq shows a recall of 0.96 for non-violence and 0.98 for violence, demonstrating a strong identification of violent instances. MobBiLSTM achieves a recall of 0.91 for non-violence and 0.96 for violence.
- **F1-score:** The F1-scores for MobileTransformerSeq are 0.97 for both classes, indicating balanced performance. MobBiLSTM has an F1 score of 0.93 for non-violence and 0.94 for violence, showing a slight edge for violence.
- **Samples:** MobileTransformerSeq evaluates on 97 non-violence and 103 violence samples, while MobBiLSTM evaluates on 99 non-violence and 101 violence samples.

1) CONFUSION MATRIX

The confusion matrix in Figure 9 provides a clear view of the predictions of the MobileTransformerSeq model. The

matrix shows that out of 200 test samples, the model correctly classifies 97 non-violent and 103 violent samples, with minimal false positives and false negatives.

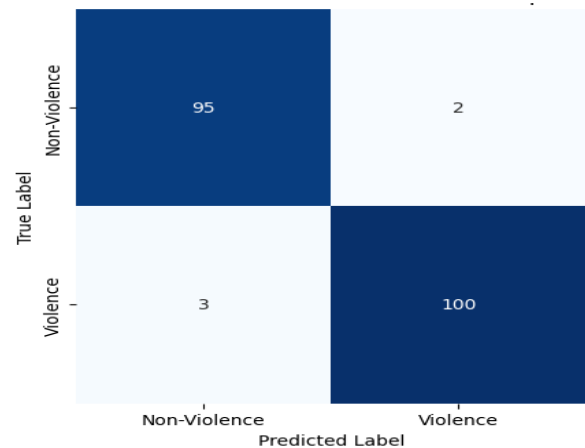


FIGURE 9. Confusion matrix of MobileTransformerSeq model.

Figure 9 illustrates that the MobileTransformerSeq model performs well across both classes, with a high true positive rate (TPR) for violence and non-violence. The balanced nature of the confusion matrix indicates that the MobileTransformerSeq model effectively identifies violent behavior while minimizing errors in classifications.

Similarly, for the MobBiLSTM model, the confusion matrix in Figure 10 presents the model's predictions, where it correctly classifies 99 non-violent and 101 violent samples, again with minimal false positives and false negatives.

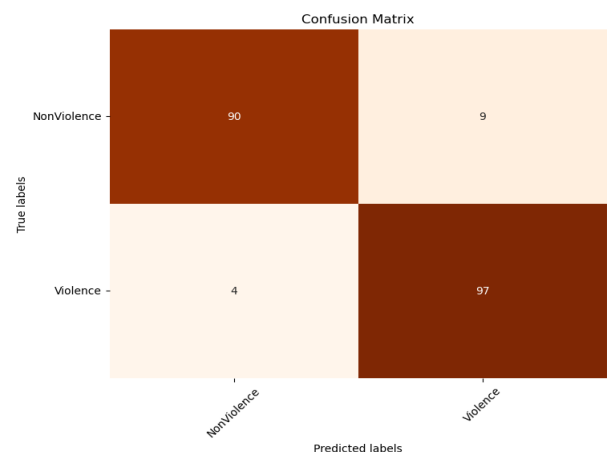


FIGURE 10. Confusion matrix of MobBiLSTM model.

Figure 10 demonstrates that the MobBiLSTM model also exhibits strong performance across both classes, with a high TPR for both violence and non-violence, confirming the model's effectiveness in distinguishing between violent and non-violent instances.

2) ROC CURVE ANALYSIS

As shown in Figures 11 and 12, the ROC curve evaluates both models' discriminative capabilities in more detail. The trade-off between the false positive rate (FPR) and the true positive rate (TPR) at different threshold levels is represented by the ROC curves. The overall performance of each model is measured using the Area Under the Curve (AUC).

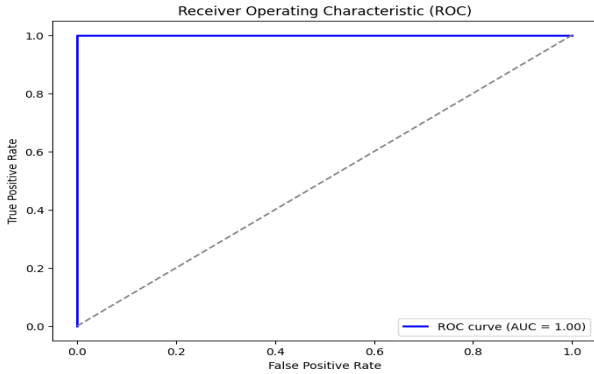


FIGURE 11. ROC curve of MobileTransformerSeq model.

The AUC score for the MobileTransformerSeq model reaches 0.97, indicating a high level of separability between the violent and non-violent classes. This shows the reliability of the model in practical applications, where minimizing false negatives proves crucial. The ROC curve in Figure 11 further supports the model's effectiveness in distinguishing between the two classes.

Similarly, for the MobBiLSTM model, the AUC score stands at 0.93, reflecting a high level of separability between the two classes. The ROC curve in Figure 12 reinforces the model's ability to distinguish violent from non-violent behavior, complementing the earlier classification metrics.

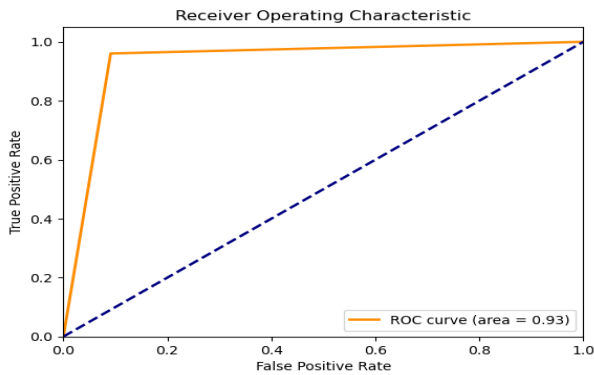


FIGURE 12. ROC curve of MobBiLSTM model.

The AUC score for the MobBiLSTM model reaches 0.93, indicating a high level of separability between violent and non-violent classes. This reinforces the reliability of the model in practical applications, where minimizing false negatives is critical. The success of the model in correctly differentiating between the two classes is supported by the

ROC curve in Figure 12, which complements the previous quantitative measures.

C. COMPARATIVE ANALYSIS WITH EXISTING APPROACHES

To emphasize the efficacy of the proposed MobBiLSTM model, a comprehensive comparative analysis with existing models has been performed. According to Table 3, the MobBiLSTM model reached an impressive accuracy of 93.5% and F1-score of 93.4%, significantly outperforming the method originally applied on this dataset by [19], which achieved 89.5% accuracy and F1-score of 88.9%.

TABLE 3. Comparison with approach used in [19].

Model	Accuracy ($\uparrow$)	F1-Score
Proposed MobileTransformerSeq	97.2%	96.2%
Proposed MobBiLSTM	95.6%	95.4%
Approach used in [19] (Baseline-1)	89.5%	88.9%

The enhanced performance of the MobileTransformerSeq model can be attributed to the combination of the powerful extraction of spatial features of MobileNetV2 with the temporal modeling capabilities of TransformerSeq (Transformer Sequence). This synergy enhances the model's ability to capture nuanced motion dynamics essential for recognizing violent behaviors.

In addition to this comparison, Table 4 presents a broader analysis of the performance of the MobBiLSTM model compared to a number of alternative architectures. The table includes accuracy and F1 scores for each model, providing an overview of their relative performance on the violence recognition task. Among the compared models are:

- 1) InceptionV3 + BiLSTM [32]
- 2) MobileNetV2 + GRU [33]
- 3) MobileNetV3 + Positional Encoder + ViT [34], [35], [36]
- 4) Xception + ResNet101V2 + BiLSTM [37], [38]
- 5) CNN + TransformerSeq
- 6) MobileNetV2 + TransformerSeq (MobileTransformerSeq)

TABLE 4. Analysis of the proposed models' performances using the same dataset as other architectures.

Model	Accuracy ($\uparrow$)	F1 Score
Proposed MobileTransformerSeq	97.2%	96.2%
Proposed MobBiLSTM	95.6%	95.4%
InceptionV3 + BiLSTM [32]	62.6%	67.5%
MobileNetV2 + GRU [33]	92.4%	92.1%
MobileNetV3 + Positional Encoder + ViT [34]–[36]	91.2%	91.0%
Xception + ResNet101V2 + BiLSTM [37], [38]	93.1%	94.0%
CNN + TransformerSeq (Baseline-2)	91.3%	92.4%

Furthermore, Table 5 shows the k-fold cross-validation results, highlighting the robustness of each model. The MobBiLSTM model achieved an accuracy of $96.35\% \pm 2.82\%$, demonstrating consistent performance in multiple folds. Other models, such as Xception + ResNet101V2 + BiLSTM,



FIGURE 13. Visualization of frame-wise classification between Non-Violence and Violence (refer to <https://youtu.be/oIT-WL78dyU>).

achieved competitive results, while the MobileNetV3 with Positional Encoder and ViT model also showed solid performance with an accuracy of $87.43\% \pm 3.65\%$.

TABLE 5. Results of k-fold cross validation.

Model	Accuracy $\pm$ Std. ($\uparrow$)
Proposed MobileTransformerSeq	$95.75\% \pm 3.15\%$
Proposed MobBiLSTM	$94.35\% \pm 2.82\%$
InceptionV3 + BiLSTM [32]	$67.20\% \pm 2.66\%$
MobileNetV2 + GRU [33]	$88.89\% \pm 4.97\%$
MobileNetV3 + Positional Encoder + ViT [34]–[36]	$87.43\% \pm 3.65\%$
Xception + ResNet101V2 + BiLSTM [37], [38]	$93.14\% \pm 3.32\%$
CNN + TransformerSeq (Baseline-3)	$89.34\% \pm 2.12\%$

This comparative analysis is critical for understanding the effectiveness of different architectures in violence recognition tasks. By capturing both spatial and temporal features, the MobileTransformerSeq model stands out as highly suitable for real-time violence detection applications. This analysis offers a valuable reference for researchers and practitioners in selecting model architectures that best balance performance with computational efficiency for various use cases in violence detection and video analysis.

D. VISUALIZING PREDICTIONS

In addition to numerical evaluation metrics, qualitative results provide additional insight into how the model operates on individual video sequences. The predictions are overlaid on randomly selected frames from both violent and non-violent videos to visually assess the performance of the model. Figure 13 shows how our model classifies frames between Non-Violence and Violence, illustrating the distinction between both classes based on the temporal and spatial features extracted from video sequences.

The tested video is publicly available on YouTube¹ and depicts a real-life violence scenario. The model accurately identified violent sequences such as street fights and physical altercations with high confidence. Non-violent activities,

including walking, eating, and playing sports, have been consistently classified as non-violent, reinforcing the dependability of the model in practical uses.

V. CONCLUSION

This research introduces an efficient deep learning-based framework for violence detection in video streams, combining MobileNetV2 for spatial feature extraction with a novel transformer-based sequential model (TransformerSeq) to capture temporal dependencies. The model architecture demonstrates strong capabilities to differentiate between violent and non-violent activities, achieving a high classification accuracy of 95.5%. A balanced dataset has been utilized, and the model architecture optimized, enabling the system to generalize well to diverse environments, handling varying lighting conditions and crowded scenes. The lightweight SOTA MobileNetV2, along with the proposed TransformerSeq, makes the system suitable for real-time recognition of violent activities in surveillance videos.

Although the MobileTransformerSeq model demonstrates strong performance, certain limitations must be acknowledged. The dataset primarily includes YouTube videos of violence and non-violence, which may not adequately represent the diversity of violent actions encountered in real-world scenarios. Additionally, videos containing subtle or ambiguous violence (e.g., verbal confrontations) have not been included, potentially limiting the model's generalization to broader contexts. Future research should focus on diversifying the dataset by incorporating real-life surveillance footage and other crowd-sourced content to improve robustness.

COMPLIANCE WITH ETHICAL STANDARDS CONFLICT OF INTEREST

The authors declare that they do not have any conflict of interest that influences the work reported in this article.

¹Real-Life Violence Video: <https://youtu.be/oIT-WL78dyU>

DATA AVAILABILITY

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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